

MYRRHA QPLANT Cryogenic Leak Rate Dashboard — Handover Package

Version: 4.0.0 | **Codename:** Critical Correction — 3× FSD575, SSoT Configuration
Date: 2026-05-10 | **Branch:** `main` | **Commit:** `ff01ec8`

1 — Project Overview

This repository packages the **MYRRHA QPLANT cryogenic helium leak-rate engineering analysis** into an idempotent, self-documenting handover system. It combines:

- **Requirements traceability** (RTM-047/048/067, EN 13185, PED 2014/68/EU)
- **Deterministic build workflows** (`setup.sh` → `build.sh` → `validate.sh` → `package.sh`)
- **Machine-readable state** (`docs/manifest.json` , `VERSION.json` , `OUTPUT_MANIFEST.json`)
- **Human-readable navigation** (40-slide master navigator, stakeholder presentation, Plotly dashboards)

v4.0.0 Key Changes

Area	Change	Impact
HP Compressors	Count corrected 4 → 3 (Kaeser FSD 575 SFC)	CAPEX −€200k
Motor power	400 kW generic → 315 kW (ALaT vendor data)	Energy models corrected
HP outlet pressure	12 barg → 14 barg nominal	Pressure maps updated
HCC inlet	1000 mbar → 1050 mbar	Air-ingress prevention
VLP suction	Fixed 300 → 400 mbar nominal (VFD 250–550)	Monte Carlo ranges updated
SSoT	New <code>data/config.yaml</code> + <code>config_loader.py</code>	No hardcoded values
Financial model	CAPEX €1.42M, ROI 1.2 yr, NPV €15.1M, IRR 83%	All slides updated

2 — Directory Structure

```

cryo_leak_rate_dashboard/
├── assets/                                # Shared CSS & JS for HTML pages
│   ├── style.css                         # Dashboard styling (responsive + print)
│   └── triage.js                         # Interactive controls (mode switch, expand/collapse)
├── data/                                 # SSoT configuration and engineering data
│   ├── config.yaml                      # ★ Single Source of Truth (human-editable)
│   ├── config.json                     # Auto-generated machine-readable SSoT
│   ├── config.md                       # Auto-generated parameter tables
│   ├── compressor_specs.json
│   ├── helium_properties.json
│   ├── leak_classes.json
│   ├── scenarios.json
│   ├── valve_candidates.json
│   ├── source_anchors.json
│   ├── standards_compliance.json
│   ├── material_costs.json
│   ├── meca_inox_specs.json
│   ├── operating_conditions.json
│   └── swagelok_specs.json
├── docs/                                # Generated documentation & visualizations
│   ├── index.html                      # Landing page (phase cards, status pills)
│   ├── index_v4.0.html                 # 40-slide master navigator (v4.0.0)
│   ├── STAKEHOLDER_PRESENTATION.html # 40-slide stakeholder deck
│   ├── manifest.json                   # Build state, file hashes, dependencies
│   ├── backlog.json                    # Tracked backlog items
│   ├── changelog.md                    # Human-readable changelog
│   ├── dashboard.html                  # Interactive Plotly dashboard
│   ├── calculations.html                # Calculation detail pages
│   ├── executive_summary.html
│   ├── rtm_traceability.html
│   ├── handover.html / .pdf            # Formal handover document
│   ├── plots/                          # 5 core Plotly plots (HTML)
│   ├── visualizations/                 # 18 v3.0 charts + backing CSV data
│   ├── visualizations_v3/              # 22 v3.1 charts (Plotly interactive)
│   ├── compressors/                    # HP redundancy & WCS.HP protection pages
│   ├── liquid_he/                      # Liquid operations guide
│   ├── standards/                      # Compliance matrix, FAT/SAT, PED
│   ├── statistical/                    # Monte Carlo, PCA, sensitivity results
│   ├── tables/                         # HTML data tables
│   ├── tables_v3/                      # Extended CSV data tables
│   └── data_exports/                   # Full conversion grids (CSV/JSON)
├── src/                                # Python source modules
│   ├── calc_leak_rate.py                # ★ Core physics engine (first-principles)
│   ├── config_loader.py                 # SSoT config loader (dot-notation access)
│   ├── generate_dashboard.py            # Main dashboard/plot generator
│   ├── build_all.py                    # Orchestrator: dashboard + handover + manifest
│   ├── build_dashboard.py               # Landing page (index.html) generator
│   ├── build_handover.py                # Handover document generator
│   ├── manifest.py                     # SHA256 hashing & idempotent writes
│   ├── compressor_reliability.py        # k-of-M availability, VFD savings
│   ├── liquid_he_loss.py                # Leak rate → liquid inventory loss
│   ├── wcs_scenarios.py                 # WCS.HP protection logic
│   ├── generate_visuals_v3.py           # v3.1 visualization generator
│   ├── generate_standards_stats.py      # Standards & statistics
│   ├── materials_db.py                  # Material properties database
│   ├── monte_carlo.py                   # Monte Carlo simulation engine
│   ├── risk_model.py                    # Risk assessment model
│   └── verify_triage.py                 # Triage compliance checker
└── tests/                              # pytest test suite (22 tests)
    ├── test_engineering.py              # Core physics validation
    ├── test_calculations.py             # Calculation integration tests
    └── test_config_loader.py            # SSoT validation tests

```


3.2 Engineering Modules

File	Purpose
src/calc_leak_rate.py	Core physics: mbar·L/s → mass flow (first-principles)
src/compressor_reliability.py	k-of-M availability, VFD energy savings
src/liquid_he_loss.py	Leak rate → liquid He inventory loss
src/wcs_scenarios.py	WCS.HP supply protection & interlocks
src/monte_carlo.py	Monte Carlo simulation engine
src/risk_model.py	Risk assessment model
src/materials_db.py	Material properties database

3.3 Visualizations

Location	Count	Description
docs/plots/	5	Core Plotly plots (leak vs loss, temp, cost, fleet, reliability)
docs/visualizations/	18	v3.0 material/supplier charts with CSV data
docs/visualizations_v3/	22	v3.1 extended charts (Monte Carlo, PCA, compressor, liquid He)

3.4 Documentation

File	Description
docs/STAKEHOLDER_PRESENTATION.html	40-slide stakeholder deck
docs/index_v4_0.html	40-slide master navigator (v4.0.0)
docs/handover.html / .pdf	Formal handover document
docs/HANDOVER_AGENT.md / .docx / .pdf	Agent-specific handover
docs/compressors/*.html	HP redundancy & WCS.HP protection
docs/liquid_he/*.html	Liquid He operations guide
docs/standards/*.md / .csv / .json	Compliance matrix, FAT/SAT, PED

3.5 Testing

File	Tests	Status
tests/test_engineering.py	5	✓ Pass
tests/test_calculations.py	varies	✓ Pass
tests/test_config_loader.py	varies	✓ Pass
tests/ test_data_integrity.py	varies	✓ Pass
tests/test_build_outputs.py	varies	✓ Pass
tests/test_outputs.py	varies	✓ Pass
Total	22	✓ All pass

3.6 Build & CI

File	Purpose
<code>setup.sh</code>	Create venv, install deps, initial build
<code>build.sh</code>	Full rebuild (dashboard + handover + manifest)
<code>validate.sh</code>	Run tests + verify outputs
<code>package.sh</code>	Create <code>dist/handover.zip</code>
<code>Makefile</code>	Make targets for all scripts
<code>.github/workflows/build.yml</code>	GitHub Actions CI pipeline

4 — Version Control Status

Property	Value
Branch	<code>main</code>
Commit	<code>ff01ec8</code>
Tag	<code>v4.0.0</code>
Build status	verified
Build timestamp	2026-05-12T09:23:12 UTC
Python	3.11.6
Key deps	numpy 1.26.4, pandas 2.2.3, plotly 5.24.1, scipy 1.14.1

Changelog Summary (v4.0.0)

- **BREAKING:** HP compressor count corrected 4 → 3 (Kaeser FSD575 SFC)
- **BREAKING:** Financial model updated (CAPEX €1.42M, ROI 1.2 yr)
- **Added:** Single Source of Truth (`data/config.yaml` + `config_loader.py`)
- **Changed:** All pressure parameters corrected from vendor/contractor data
- **Changed:** Monte Carlo distributions updated (VLP, LP ranges)
- **Fixed:** FSD575 power specs, energy consumption, cost models
- **Validated:** 22/22 tests passing, all outputs verified

5 — Quick Start for New Agents

Step 1: Read state files

```
cat VERSION           # → 4.0.0
cat docs/manifest.json # Build state, file hashes
cat docs/backlog.json  # Open items
cat CHANGELOG.md       # Version history
```

Step 2: Setup environment

```
./setup.sh           # Creates venv, installs deps, runs initial build
```

Step 3: Validate

```
./validate.sh        # Runs 22 tests, checks outputs
```

Step 4: Build (if needed)

```
./build.sh           # Full rebuild of all artifacts
```

Step 5: Package (for distribution)

```
./package.sh         # Creates dist/handover.zip
```

Key files to read first

1. `data/config.yaml` — All engineering parameters (SSoT)
2. `src/calc_leak_rate.py` — Core physics engine
3. `docs/manifest.json` — Current build state
4. `CHANGELOG.md` — What changed and why

6 — Idempotent Rebuild Instructions

The entire project can be rebuilt from source files:

```
# 1. Fresh setup
./setup.sh

# 2. Full build (generates all HTML, plots, tables, manifests)
./build.sh

# 3. Run tests and verify outputs
./validate.sh

# 4. Package for distribution
./package.sh
```


Idempotency guarantee: Running `build.sh` multiple times produces identical outputs. File writes only occur when content has changed (SHA256-based).

Source files (inputs to build)

- `data/*.json` , `data/*.yaml` — Engineering data
- `src/*.py` — Calculation and generation modules
- `assets/style.css` , `assets/triage.js` — Web assets
- `VERSION` , `VERSION.json` — Version metadata

Generated files (outputs of build)

- `docs/*.html` — All HTML pages
- `docs/plots/` , `docs/visualizations*/` — All charts
- `docs/manifest.json` — Build manifest
- `outputs/` — Calculation results
- `dist/` — Distribution package

7 — Companion Files

File	Purpose
<code>ARTIFACTS.yaml</code>	Machine-readable catalog of all deliverables
<code>DOWNLOAD_CHECKLIST.md</code>	What to download for full handover
<code>REBUILD_GUIDE.md</code>	Detailed rebuild instructions with troubleshooting